Chapter 8 — Chemical Equilibrium

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. At equilibrium, concentrations of reactants and products are: (a) Zero (b) Increasing (c) Constant (d) Equal

2. Kc expression uses: (a) Volumes only (b) Masses only (c) Initial concentrations (d) Equilibrium concentrations

3. Pure solids in Kc are: (a) Added (b) Inverted (c) Squared (d) Omitted

4. K changes with: (a) Temperature (b) Concentration (c) Catalyst (d) Pressure

5. Catalyst at equilibrium: (a) Changes K (b) Removes product (c) Stops reverse (d) Speeds both directions

6. Higher pressure favours: (a) More gas moles (b) Solid only (c) Fewer gas moles (d) Equal gas moles

7. For A(g) + 3B(g) ⇌ 2C(g), high pressure favours: (a) C (b) A (c) No change (d) B

8. Kp expression uses: (a) Volumes (b) Densities (c) Molar masses (d) Partial pressures

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why do concentrations remain constant at equilibrium?

2. Why are pure solids omitted from equilibrium expressions?

3. How does the magnitude of K indicate the position of equilibrium?

4. Why does a catalyst not change K?

5. How does temperature affect an exothermic equilibrium?

6. How does removing a product affect a reversible reaction?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the equilibrium constant expression and its units for gaseous reactions. **(4)**

(b) State characteristics of K and explain Le Chatelier’s principle. **(4)**

Q.4 (a) Explain effects of concentration, pressure and temperature on equilibrium. **(4)**

(b) Write Kc and Kp expressions and explain why a catalyst does not change equilibrium yield. **(4)**

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